

ACID, BASE & SALT

- **Acid**
- It is derived from "Acidus".

Acidus [Latin word]

means → Sour to taste

For example:-

- ✓ Taramind, lemon juice, tea, curd, orange, guava, green apple, sweat, raw mango etc.

- **Some substances containing acid**

S.No	Substance	Contain acid	M.F
1.	Critic fruits, orange juice, Lemon juice	Critic acid	$C_6H_8O_7$
2.	Green apple	Malic acid	$C_4H_6O_5$
3.	Sweat	Lactic acid	$C_3H_6O_3$
4.	Curd	Lactic acid	$C_3H_6O_3$
5.	Sour milk	Lactic acid	$C_3H_6O_3$
6.	Taramind (emli)	Tartaric acid	$C_4H_6O_3$
7.	Grapes	Tartaric acid	$C_4H_6O_6$
8.	Vinegar (Sirka)	Acetic acid	CH_3COOH
9.	Tomato	Oxalic acid	$C_2H_2O_4$
10.	Guava	Oxalic acid	$C_2H_2O_4$
11.	Ant sting (Red ant), Bee, Nettle sting	Formic acid OR Methanoic acid	$HCOOH$
12.	Cold drink	Carbonic acid	H_2CO_3
13.	Tea	Tanic acid	$C_{76}H_{52}O_{46}$
14.	Amla	Ascorbic acid (Vitamin - C)	$C_6H_8O_6$
15.	All fruits and vegetables	Natural acid (Weak acid)	

Note:

1. Coffee and black tea contain acid but green tea contain base.
2. Coffee contains mainly chlorogenic and quinic acids.

- **Characteristics of acid**

1. Sour to taste.
2. Mostly acids contain Hydrogen atom in their molecular formula.
3. High solvent power.
4. Having capacity to convert one dye to another dye, except - red.
5. It accepts lone pair of electrons.
6. It turns blue litmus paper to red.
7. Evolution of hydrogen gas when it reacts with metal.

$$\begin{array}{lclclcl} \text{high reactive metal} & + & \text{Acid} & \longrightarrow & \text{Salt} & + & \text{Hydrogen } (\uparrow) \\ 2\text{Na} & + & 2\text{HCl} & \longrightarrow & 2\text{NaCl} & + & \text{H}_2 (\uparrow) \\ \text{Mg} & + & 2\text{HCl} & \longrightarrow & \text{MgCl}_2 & + & \text{H}_2 (\uparrow) \\ \text{Zn} & + & \text{H}_2\text{SO}_4 (\text{diluted}) & \longrightarrow & \text{ZnSO}_4 & + & \text{H}_2 (\uparrow) \end{array}$$
8. It gives CO_2 when reacts with metal carbonate and metal Hydrogen Carbonate.

i.e,



For example:-



9. When it dissolves in water it forms a colourless solution.

10. It changes methyl orange to red.

11. It strongly reacts with base to form salt and water.

i.e,

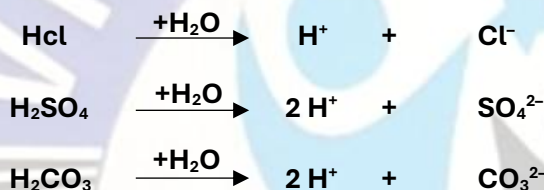


For example:-



12. It conducts electricity because the aqueous solution of acid contain Hydrogen ion (H^+).

13. The aqueous solution of acid gives H^+ ions.

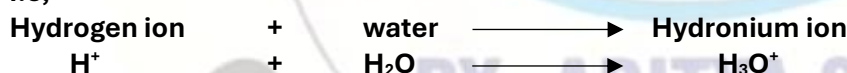


14. It forms hydronium ion when reacts (dissolve) with water.

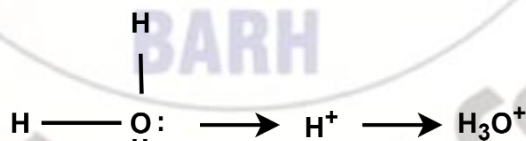


We know that the aqueous solution of acid gives H^+ ions & this H^+ ion reacts with water to form hydronium ions.

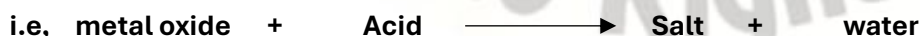
i.e,



Chemically bond formation



15. Reaction of metallic oxide with acids



For example:-



↑
Metallic oxides are basic in nature

16. It accepts protons.

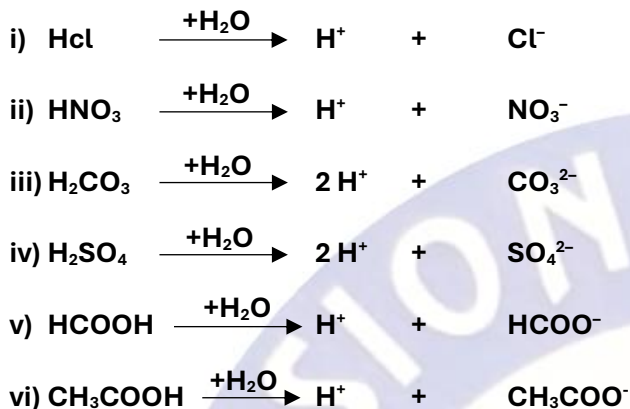
• **Definition of an acid a/c to Arrhenius**

- An acid is that substance which gives hydrogen ion (H^+) when it dissolves in water.

In other hand

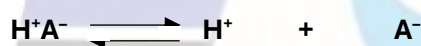
“The aqueous solution of an acid gives H^+ ions”

For example:-



▪ **Classification of an acid**

Case 1: On the basis of strength



$$K_a = \frac{[H^+] + [A^-]}{[H^+ A^-]}$$

Acidity Concentration of an acid

Similarly,

$$p^H = -\log [H^+]$$

Power of concentration of H^+

$$\Rightarrow p^H = -1 \times \log [H^+]$$

$$\Rightarrow p^H = \log [H^+]^{-1} \quad [\because n \log m = \log m^n]$$

$$\Rightarrow p^H = \log \frac{1}{[H^+]} \quad [\because \log m^{-1} = \log \frac{1}{m}]$$

Logarithmic function

i.e, $p^H \propto \frac{1}{[H^+]}$

Similarly,

$$\Rightarrow p^{K_a} = -\log K_a$$

$$\Rightarrow p^{K_a} = \log (K_a)^{-1}$$

$$\Rightarrow p^{K_a} = \log \frac{1}{K_a}$$

i.e, $p^{K_a} \propto \frac{1}{K_a}$

1. Strong acids

- Those acids which dissolved in water get almost completely dissociated to provide large amount of hydrogen ions. These type of acids are called strong acids.

→ **In strong acids:-**

$[H^+]$	(↑)	p^H	(↓)
Ka	(↑)	p^{Ka}	(↓)

For example:-

- ✓ HCl → Hydrochloric acid
- ✓ H_2SO_4 → Sulphuric acid
- ✓ HNO_3 → Nitric acid
- ✓ HBr → Hydrogen Bromide
- ✓ HF → Hydrogen Fluoride, etc.

2. Weak acids

- Those acids which dissolved in water, get partially dissociated to provide less amount of hydrogen ions. These acids are called weak acids.

→ **In weak acids:-**

$[H^+]$	(↓)	p^H	(↑)
Ka	(↓)	p^{Ka}	(↑)

For example:-

- ✓ H_2NO_2 → Nitrous acid
- ✓ H_2CO_3 → Carbonic acid
- ✓ CH_3COOH → Acetic acid
- ✓ H_3BO_3 → Boric acid
- ✓ H_2SO_3 → Sulphurous acid
- ✓ HNO_2 → Nitrous acid

- All foods items having weak acids.
- Weak acids are called organic acids and natural acids.

Case 2: On the basis of Carbon – atom

1. Organic acids

- Those acids which contain Hydrogen & Carbon atoms are called organic acids.
- All food items are used in everyday life having organic acids.

For example:-

- ✓ $HCOOH$ → Formic/methanoic acid
- ✓ CH_3COOH → Acetic/Ethanoic acid

- These are actually hydrocarbons.

exception

- ✓ H_2CO_3 → Carbonic acid is not organic acid

Note:

- Organic acids can be prepared in laboratory by simple process.

2. Inorganic/mineral acids

- Those acids which do not contain carbon atoms are called inorganic acids.
- These are generally used in laboratory.

- These are strong acid.
- These are obtain from minerals.

For example:-

- ✓ H_2SO_4 —————> Sulphuric acid
- ✓ H_3PO_4 —————> Phosphoric acid
- ✓ HNO_3 —————> Nitric acid
- ✓ HNO_2 —————> Nitrous acid
- ✓ HCl —————> Hydrochloric acid/Gastric acid
- ✓ HI —————> Hydrogen iodide

Note:

- Mineral acids are widely used in industrial point of view. These are used in the manufacture of fertilizer industry, chemical, explosives dyes, pigments etc.

Case 3: On the basis of amount of acid in the solution

1. Concentrated acid

- A solution containing the maximum amount of acid is called concentrated acid.

For example:-

- ✓ Concentrated HCl —————> $[\text{HCl}]$
- ✓ Concentrated CH_3COOH —————> $[\text{CH}_3\text{COOH}]$

2. Diluted acid

- A solution containing minimum or less amount of acid is called diluted acid. i.e, A solution in which the amount of water is more than amount of acid.

For example:-

- ✓ Gastric juice (Diluted HCl)
- ✓ H_2CO_3 Diluted Carbonic acid
- ✓ CH_3COOH Diluted Acetic acid
- ✓ H_3BO_3 Diluted Boric acid

Note:

- Diluted acid can be prepared by adding water in concentrated acid.

Case 4: On the basis of element present

1. Oxo acids

- Those acids which contain oxygen as well as hydrogen are called oxo acids.
- Hydrogen must be attached with Oxygen.
- Oxo acids are also called Oxy acid and ternary acids.

For example:-

- ✓ H_2SO_4 —————> Sulphuric acid
- ✓ H_3PO_4 —————> Phosphoric acid
- ✓ H_3PO_3 —————> Phosphorus/Orthophosphorus acid
- ✓ H_2PO_2 —————> Hypophosphorus acid
- ✓ HNO_2 —————> Nitrous acid
- ✓ HNO_3 —————> Nitric acid

2. Hydra acids

- Those acids which contain only hydrogen atoms; not contain oxygen atoms are called Hydra acids.

For example:-

- ✓ HCl —————> Hydrogen chloride
- ✓ HI —————> Hydrogen iodide
- ✓ HF —————> Hydrogen Fluoride
- ✓ HBr —————> Hydrogen Bromide
- ✓ HCN —————> Hydrogen Cyanide
- ✓ H₂S —————> Hydrogen Sulphide

Case 5: On the basis of “basicity of an acid” of proticity of an acid.

▪ Basicity/Proticity of an acid

- The number of replaceable hydrogen atom present in a molecule of an acid is called basicity of an acid.

▪ Replaceable Hydrogen atoms

- That hydrogen atom which are displaced by metal or group of elements are called replaceable hydrogen.

1. Monobasic/Manoprotic acids

- Those acids which contain one replaceable hydrogen atom are called monobasic acids.

For example:-

- ✓ HCl, HI, HBr, HCN, HF etc.

2. Dibasic/Diprotic acids

- Those acids which contain two replaceable hydrogen atoms are called dibasic acids.

For example:-

- ✓ H₂SO₄, H₂SO₃, H₂CO₃, H₂S etc.

3. Tribasic/Triprotic acids

- Those acids which contain three replaceable hydrogen atoms are called tribasic acids.

For example:-

- ✓ H₃PO₄ —————> Phosphoric acid
- ✓ H₃PO₃ —————> Phosphorus acid

Note:

- Boric acid (H₃BO₃) is a monobasic acid.

4. Tetrabasic/Tetraprotic acids

- Those acids which contain four replaceable hydrogen atoms are called tetrabasic acids.

For example:-

- ✓ H₄P₂O₇ —————> Pyrophosphoric acid

Acid	No. of Replaceable Hydrogen gas	Basicity
HCl	1	1
H ₂ SO ₄	2	2
H ₃ PO ₄	3	3
H ₄ P ₂ O ₇	4	4

- **Some acids and its uses**

1. **Hydrochloric acid (HCl)**

- In the textile industry for dyeing.
- For extracting glue from bones.
- For cleaning iron sheets before galvanization.
- For cleaning tiles, toilets etc.
- For in P.V.C (polyvinyl chloride).
- Digestion of food (gastric juice).

2. **Nitric acid (HNO₃)**

- In the refining gold & silver.
- For preparing fertilizer, explosive dyes and drugs etc.
- Making in ammonium nitrate.
- Explosive items – TNT, Dynamite.

3. **Sulphuric acid (H₂SO₄)**

- For preparing fertilizers, detergents, synthesis fibre, plastics.
- In petroleum industry for refining.
- Cleaning metal surface before electroplating.
- In lead acid batteries.
- In making HCl and alum.

4. **Boric acid (H₃BO₃)**

- Widely used in making detergent.
- As a food preservative.
- In the manufacture of glass, leather, paper, adhesive and explosive.

5. **Phosphoric acid (H₃PO₄)**

- Making in fertilizers and detergent industry.

6. **Citric acid (C₆H₈O₇)**

- As a preservation of food.
- As a flavouring agent.

7. **Ascorbic acid of vitamin C (C₆H₈O₆)**

- In the treatment of bone marrow.
- In the treatment of scurvy disease.

8. **Acetic acid (CH₃COOH)**

- A vinegar to preserve food and a flavouring agent.
- Added to pickles to make them sour.

9. **Tartaric acid (C₄H₆O₆)**

- Making in baking powder.

- **The Bronsted Lowry theory**

- A/C to the Bronsted Lowry theory, A proton doner is called an acid and a proton acceptor is called base.
i.e,

Proton donor → Acid

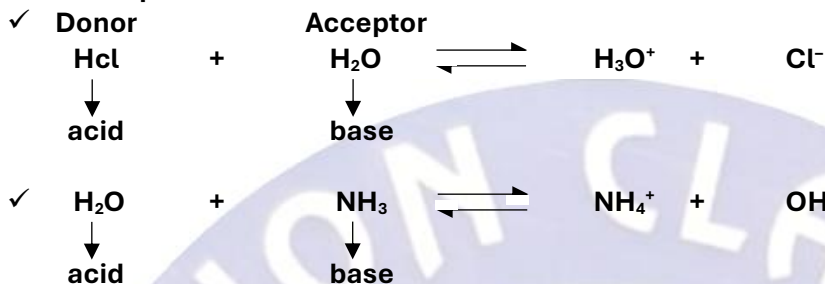
Proton acceptor → Base

- This theory is called the proton donor acceptor theory.

Note:

- H^+ is known as proton which does not exist in free state but can be transformed from one substance to another substance.

For example:-



Special Note:

- Water is not just a solvent it acts either as an acid or as a base.
 ➤ Water may or may not be a reactant.

• **Conjugate base**

- The residual part of an acid after the removal of proton is called the conjugate base of that acid.
 i.e, Acid - H^+ (Proton) → Conjugate base

For example:-



Cl⁻ is conjugate base of HCl (acid).

• **Conjugate Acid**

- The substance resulting after the addition of proton to a base is called the conjugate acid of that base.
 i.e, Base + H^+ → Conjugate acid

For example:-



• **Conjugate pair**

- The acid and its conjugate base is called conjugate pair.
 ➤ Similarly, the base and its conjugate acid is called conjugate pair.

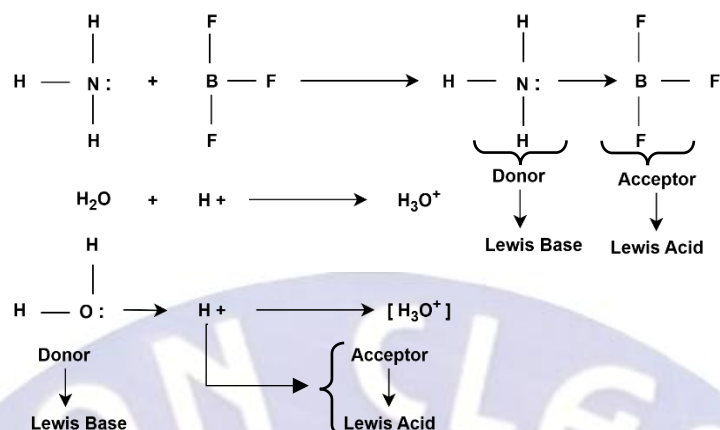
• **Lewis theory**

- This theory of acid and base was postulated by G.N Lewis in 1943.
 ➤ A/C to Lewis theory electron pair acceptors are called acid and electron pair donor are called base.

i.e, Electron pair acceptors → Lewis acid

Electron pair donor → Lewis base

For example:-



BASE

- **Base**

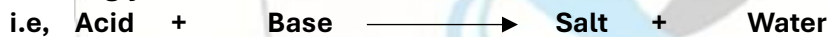
➤ A base is that substance in which is “bitter in taste”.

For example:-

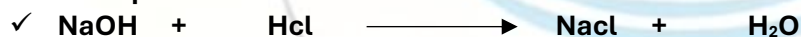
- ✓ Soap solution, detergent solution, toothpaste, caustic soda (NaOH), quick lime (CaO), washing Soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$), Baking Soda (NaHCO_3), Neem etc.

- **Characteristics of a base**

1. It is bitter in taste.
2. It is soapy/slippery to touch.
3. It strongly reacts with an acid to form salt and water.

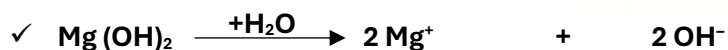
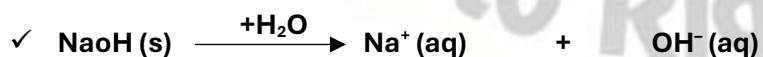


For example:-



4. It donates lone pairs of electrons.
5. It changes red litmus paper to blue.
6. It changes from methyl orange to yellow.
7. It accepts protons (H^+).
8. It has detergent action.
9. The aqueous solution of a base gives Hydroxide ions (OH^-).

For example:-

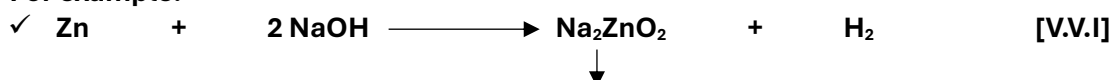


10. It neutralizes any acidic reactions.

11. Some metals react with base to form salt and Hydrogen gas.



For example:-

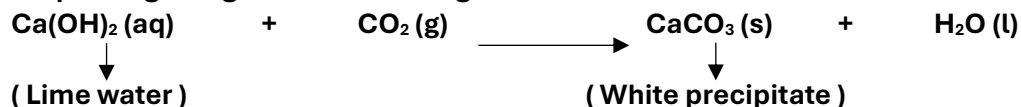


(Sodium Zincate) $\longrightarrow$ A complex salt

12. It has Hydroxide ion (OH^-) in its M.F (Molecular Formula).

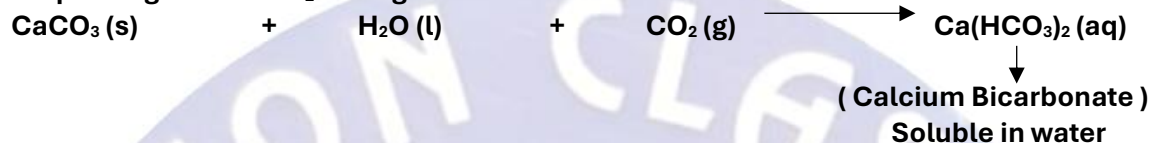
Note:

1. On passing CO_2 gas evolved through lime water.



Now,

On passing excess CO_2 through Lime water.



➤ This reaction is used to check in CO_2/CO_3 .

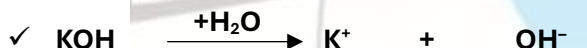
Example of bases:-

✓ LiOH , NaOH , KOH , RbOH , CsOH , FrOH , Be(OH)_2 , Mg(OH)_2 , Ca(OH)_2 , Sr(OH)_2 , etc.

• Definition of a base A/C to Arrhenius

➤ A/c to Arrhenius, A base is that substance which aqueous solution gives Hydroxide ion (OH^-).

For example:-



Note:

1. The water soluble bases are called alkali.

For example:-

✓ LiOH , NaOH , KOH , RbOH , CsOH , Fr(OH) , Mg(OH)_2 , Ca(OH)_2 , Be(OH)_2 , Sr(OH)_2 , etc.

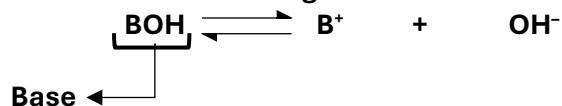
2. All alkalies are bases but all bases are not alkalies.

For example:-

✓ Al(OH)_3 , Cu(OH)_2 , etc are not alkalies.

▪ Classification of a base

Case 1: On the basis of strength



$$\underbrace{K_b}_{\substack{\text{basicity} \leftarrow \\ \text{concentration of a base}}} = \frac{[B^+] + [OH^-]}{[BOH]}$$

$$\therefore p^{OH} = -\log [OH^-]$$

$$p^{OH} = \log \frac{1}{[OH^-]}$$

$$\text{i.e., } p^{OH} \propto \frac{1}{[OH^-]}$$

Similarly,

$$p^{Kb} = -\log K_b$$

or,

$$p^{Kb} \propto \frac{1}{K_b}$$

1. Strong bases

- Those bases when dissolved in water get almost completely dissociated to provide large amount of hydroxide ions, these type of bases are called strong bases.
- In strong base: $[OH^-] (\uparrow)$ $p^{OH^-} (\downarrow)$
 $K_b (\uparrow)$ $p^{Kb} (\downarrow)$

For example:-

- ✓ NH_4OH , $Ca(OH)_2$, KOH , $NaOH$, $RbOH$ etc.

2. Weak bases

- Those bases when dissolved in water get partially dissociated to provide less amount of hydroxide ions are called weak acids.
- In weak bases: $[OH^-] (\downarrow)$ $p^{OH^-} (\uparrow)$
 $K_b (\downarrow)$ $p^{Kb} (\uparrow)$

For example:-

- ✓ $Mg(OH)_2$, $Cu(OH)_2$, $Fe(OH)_3$, $Zn(OH)_2$, $Al(OH)_3$ etc.

Case 2: On the basis of C – atom

1. Organic bases

- Those bases which having C – atoms are called organic bases.

For example:-

- ✓ Alkyl Amine, Pyridine etc.

2. Inorganic bases/mineral bases

- Those bases which have no – any C – atoms are called inorganic bases.

For example:-

- ✓ $NaOH$, KOH , $RbOH$, $Mg(OH)_2$, $Ba(OH)_2$, $Fe(OH)_2$, $Al(OH)_3$ etc,

Strong Bases

Weak Bases

Note:

- All inorganic bases are either strong or weak bases.

Case 3: On the basis of acidity of a bases

- **Acidity of a base**

- The no. of replaceable hydroxide ions present in a base is called acidity of a base.

- **Replaceable Hydroxide ions**

- That hydroxide ions which present in any base which is displaced by non – metal or metal is called replaceable hydroxide ions.

1. **Monoacidic bases**

- Those bases which have only one replaceable Hydroxide ion are called monoacidic bases.

For example:-

- ✓ KOH, NaOH, RBOH, CsOH, NH₄OH.

2. **Di – acidic bases**

- Those bases which having two replaceable hydroxide ions are called di – acidic bases.

For example:-

- ✓ Mg(OH)₂, Ca(OH)₂, Sr(OH)₂, Ba(OH)₂ etc.

3. **Triacidic bases**

- Those bases which having three replaceable hydroxide ions are called tri – acidic base.

For example:-

- ✓ Fe(OH)₃ → Ferric Hydroxide.
- ✓ Al(OH)₃ → Aluminum Hydroxide

Note:

1.

Base	Acidity
NaOH	1
Mg(OH) ₂	2
Al(OH) ₃	3

2. Water soluble bases.

For example:-

- ✓ KOH, RbOH, CsOH etc.

3. Water insoluble bases.

For example:-

- ✓ Fe(OH)₃, Al(OH)₃.

4. Oxide and Hydroxide of element of

(1 – Group)	(2 – Group)
LiOH, Li ₂ O → Oxide, Na ₂ O/NaOH, Na – peroxide, KOH, K ₂ O, RbOH, CsOH	BeO/Be(OH) ₂ , MgO/Mg(OH) ₂ , CaO/Ca(OH) ₂ , Sr(OH) ₂ /SrO, BaO/Ba(OH) ₂ , RaO/Ra(OH) ₂

- Both groups are generally basic in nature.

5.

(13 – Group)	14 th group oxide/ Hydroxide of acids
B – B ₂ O ₃ /B(OH) ₃	C – CO ₂
Al ₂ O ₃ Al	SiO ₂
Ga ₂ O ₃ Ga	GeO ₂
In ₂ O ₃ In	SnO ₂
Tl ₂ O ₃ Tl	PbO ₂

Amphoteric

- Amphoteric/Amphiprotic oxides
- Some metal oxides can act as either an acid or base in a reaction to produce a salt and water.

For example:-



➤ Those metal which having dual nature are called Amphoteric metals. (Acid as well as basic nature)

For example:-

Trick:

जॉन	अली	ने	बे	कार	गया	पंजाबी	सांग
↓	↓		↓	↓	↓	↓	↓
ZnO	Al ₂ O ₃		BeO	Cr ₂ O ₃	Ga ₂ O ₃	PbO	SnO
आज	सब	बाबा	इंदौर	गये			
↓	↓	↓	↓	↓			
As ₂ O ₃	Sb ₂ O ₃	Br ₂ O ₃	In ₂ O ₃	GeO			

- Amphoteric oxide behaves acid as well as base.
- Neutral oxide neither reacts with acid nor with base [CO, NO].

- **Uses of bases**

- 1. Sodium Hydroxide:-** **[NaOH or Caustic Soda]**
 - In soap and detergent industry.
 - For the manufacture of rayon.
 - In paper and pulp industry.
 - For manufacturing other chemicals purify the petroleum.
- 2. Calcium Hydroxide:-** **[Ca(OH)₂ or Slaked lime (famous name)]**
 - For white washing.
 - For the manufacturing of bleaching powder.
 - In leather industry.
 - For softening hardness of water.
 - For neutralizing acidity of the soil.

3. **Ammonium Hydroxide:-** $[\text{NH}_4\text{OH}]$

- As a cleaning agent.
- For preparing ammonium salt.
- Making fertilizers, plastic and dyes.

4. **Magnesium Hydroxide:-** $[\text{Mg}(\text{OH})_2]$

- Used in antacid formation.

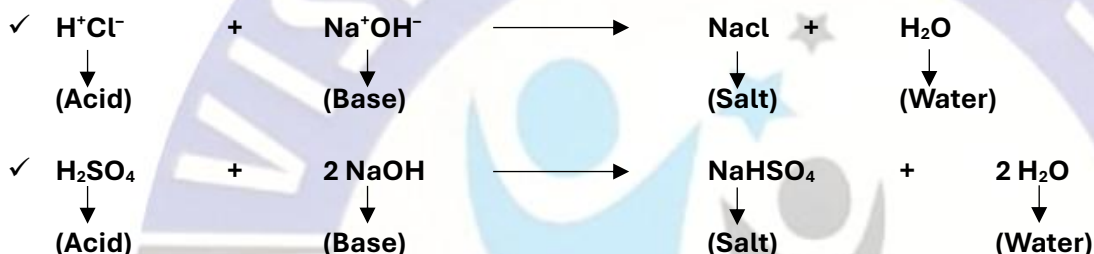
↓
Milk of magnesia (Antacid gel) reduce the acidity of stomach $\text{Mg}(\text{OH})_2$.

SALT

• **Salt**

- A salt is formed when an acid strongly reacts with base.
- In the reaction of acid and base, salt and water both are formed.

For example:-



Note:

1. In NaCl
 Na^+ is basic radical.
 Cl^- is acidic radical.
2. In NaCl , KCl , RbCl etc having same radicals chlorine (Cl) So, these are called chloride salt.

Types of Salt

1. **Common Salts**

- Those salts which have neither hydrogen ions nor Hydroxide ions are called common salts.

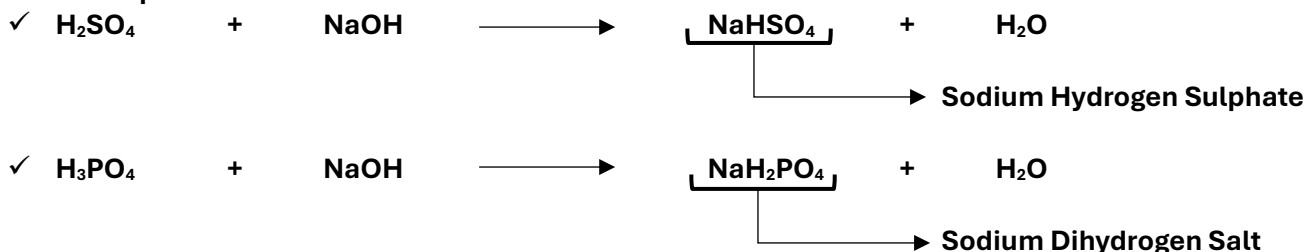
For example:-

- ✓ NaCl (Used everyday in life), KCl , Na_2SO_4 , Na_3PO_4 , KNO_3 , Na_2CO_3 , CuSO_4 etc.

2. **Acidic Salts**

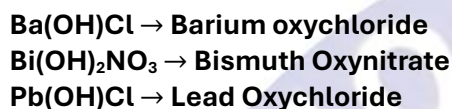
- Those salts which have replaceable hydrogen atoms are called acidic salts.

For example:-



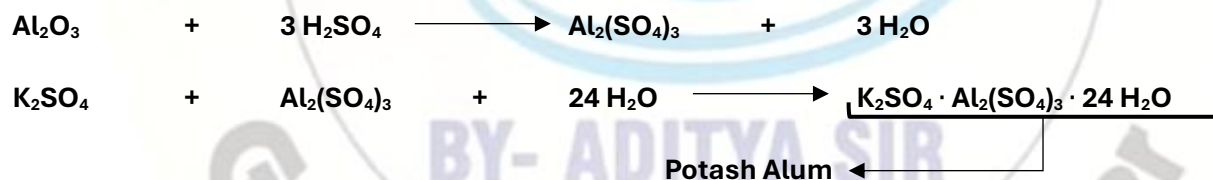


- For example:-**



4. Double Salts

- For example:-**



- **Uses of Potash Alum**

- In purifying water.
- As a antiseptic.
- In dying.
- In tearing milk.

5. Mixed Salts

- For example:-**



Note:

- **There must be more than one cation and anion for a mixed salt.**

6. Complex Salts

- Those salts which are formed by mixing of saturated solution of simple salts by crystallization of solution similar to double salts.

For example:-

- ✓ $K_4[Fe(CN)_6] \rightarrow$ Potassium ferrocyanide
- ✓ $K[Ag(CN)_2] \rightarrow$ Potassium Argento Cyanide
- ✓ $Na[Ag(CN)_2] \rightarrow$ Sodium Argento Cyanide

Difference between Double Salt & Complex Salt

	Double Salt	Complex Salt
1.	Double Salt lose their identity in solution and break down into simpler ions.	Complex Salt retain their identity in water or in other solvents.
2.	Double Salt exist in crystalline state.	Complex Salt exist in solid state as well as in solution form.

7. Hydrated Salts

- Those salts which contain water in their crystalline form, are called hydrated salts.

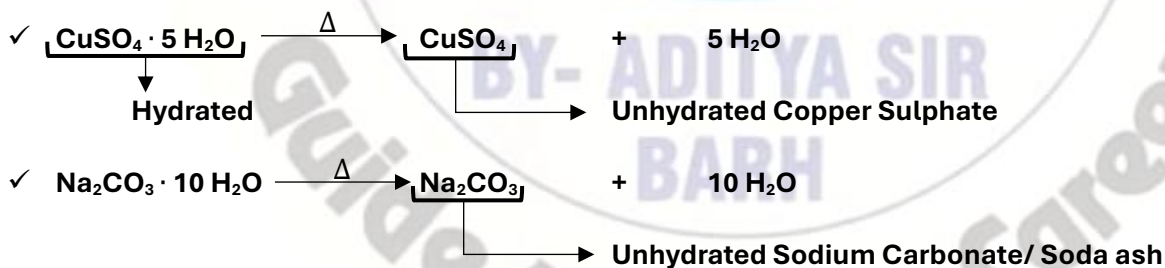
For example:-

- ✓ $CuSO_4 \cdot 5 H_2O \rightarrow$ Blue Vitriol or Copper Sulphate.
- ✓ $Na_2CO_3 \cdot 10 H_2O \rightarrow$ Washing Soda or Sodium Carbonate.
- ✓ $CaSO_4 \cdot \frac{1}{2} H_2O \rightarrow$ Plaster of Paris.
- ✓ $CaSO_4 \cdot 2 H_2O \rightarrow$ Gypsum.

8. Unhydrated Salt

- Those salt which do not contain water in their crystalline form, are called unhydrated salts.

For example:-



9. Deliquescent Salts

- Those salts which absorb water vapour from atmosphere and start sweating these type of salts are called deliquescent salt.

For example:-

- ✓ Unhydrated Calcium Chloride

10. Eno's Salts

- Those salts which are mixture of Sodium bicarbonate and citric acid.
- These are used to reducing the acidity of stomach.

11. Rock Salt

- Those salts which are obtained from minerals having red smooth soil present in the form of impurities due to this reason its colour is brown.

12. Iodised Salts

- Those salts which contain a little amount of KI or KIO_3 are mixed in common salt (NaCl) then the formed salt is known as iodised salts.

Note:

- Due to lackness of iodine in our body, people suffering by Goitre disease.

- Some important Salts and their molecular formula

Important Salts	Molecular Formula
Sodium Chloride	NaCl
Calcium Chloride	$CaCl_2$
Magesium Chloride	$MgCl_2$
Zinc Chloride	$ZnCl_2$
Sodium Sulphate	Na_2SO_4
Potassium Sulphate	K_2SO_4
Calcium Sulphate	$CaSO_4$
Magnesium Sulphate	$MgSO_4$
Aluminium Sulphate	$Al_2(SO_4)_3$
Zinc Sulphate	$ZnSO_4$
Copper Sulphate	$CuSO_4$
Ammonium Sulphate	$(NH_4)_2SO_4$
Sodium Nitrate	$NaNO_3$
Potassium Nitrate	KNO_3
Sodium Carbonate	$NaCO_3$
Calcium Carbonate	$CaCO_3$
Zinc Carbonate	$ZnCO_3$
Sodium acetate	CH_3COONa

- Indicator

- An indicator is a substance that assume different colour in acidic, basic and neutral solution.

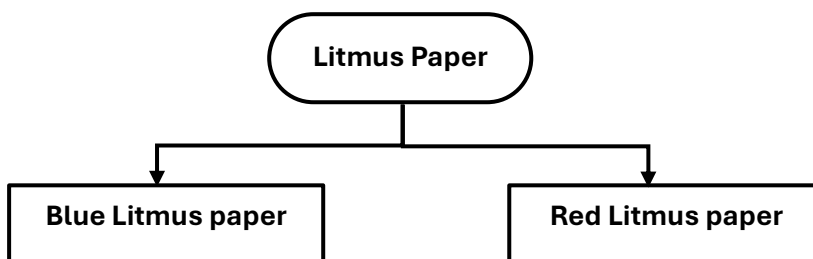
For example:-

- ✓ Litmus, Methyl Orange and phenol – phthalein are the most useful indicators.

Indicator	Acidic Solution	Basic Solution	Neutral Solution
Blue – Litmus paper	Red	No Change	No Change
Red – Litmus paper	No Change	Blue	No Change
Methyl Orange	Red	Yellow	Orange
Phenolphthalein	Colourless	Pink	Colourless

- Litmus

- It is a natural dye made from small plant called lichens. The lichens extract, called litmus Solution.
- It is neutral and purple in colour.
- The extract when acidified gives red litmus solution.
- The extract when made alkaline gives blue litmus solution.



1. Blue Litmus Paper

- Blue Litmus Paper is prepared by dipping a strip of filter paper in blue litmus solution. The paper is then removed from the solution and dried.

2. Red Litmus Paper

- Red Litmus paper is prepared by dipping a strip of filter paper in red litmus solution. The paper is then removed from the solution and dried.

Note:

- 'Lichen' belongs to the "Thallophyta division" plant.

- Phenolphthalein

- It is a colourless compound an alcoholic solution of phenolphthalein is used as an indicator.

- Methyl Orange

- A very small amount of solid methyl orange is dissolved in hot water and filtered. The filtrate is used as an indicator.

Note:

- Methyl Orange and Phenolphthalein are synthetic – indicators.

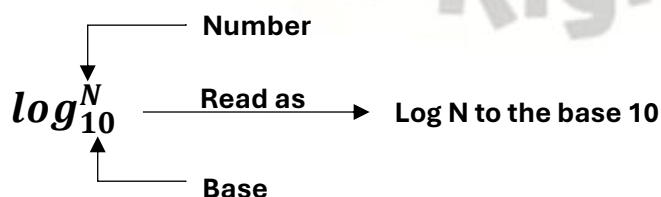
- Olfactory indicator

- Some indicators like as onion juice vanilla essence and clove oil which can be identified by a blind person through smelling.
- Olfactory indicators change their smell in acidic & basic solution.

- Household indicators

Household indicator	Acidic Solution	Basic Solution
Turmeric Juice	Yellow	Deep brown
Red Cabbage	Red	Green
Beat Root	Red Violet	Yellow

- Logarithmic function



- Properties of logarithmic function

- $\log_{10}^{10} = \log_1^1 = \log_e^e = \dots = 1.$
- $\log_{10}^{m^n} = n \log_{10}^m.$

- iii) $-\log_{10}^m = \log_{10}^{m^{-1}} = \log_{10} \left(\frac{1}{m} \right)$.
- iv) $\log_{10}^{mn} = \log m + \log n$.
- v) $\log_{10}^{-N} = \text{Can't be negative}$.
- vi) $\log_{10}^1 = 0$.

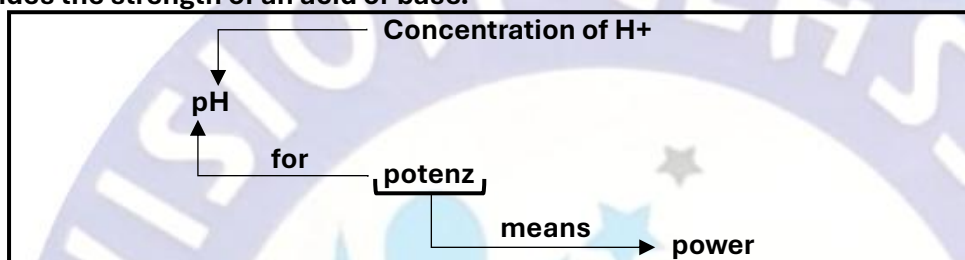
Note:

$$\log_{10}^N = \log^N$$

Base can be hide

• **pH – Scale**

- In 1909, S.P Sorenson made a special type of scale which measured the concentration of H^+ – ion called pH scale.
- It decides the strength of an acid or base.



• **Definition of pH**

- The pH of an aqueous solution is the negative logarithm of its concentration of H^+ ion.
- i.e,

$$\Rightarrow pH = -\log [H^+]$$

$$\Rightarrow pH = (-1) \log [H^+]$$

$$\Rightarrow pH = \log [H^+]^{-1}$$

$$\Rightarrow pH = \log \frac{1}{[H^+]}$$

$$[\because n \log m = \log m^n]$$

$$[x^{-1} = \frac{1}{x}]$$

$[H^+]$ → Read as concentration of H^+ .

$[OH^-]$ → Read as concentration of OH^- .

$[]$ → Symbol of concentration.

Note:

1. $pH = \log \frac{1}{[H^+]}$

i.e, $pH \propto \frac{1}{[H^+]}$

$\therefore [H^+] \uparrow$, $pH \downarrow$ & $[H^+] \downarrow$, $pH \uparrow$

Similarly,

pOH

power of concentration of $[OH^-]$

$\therefore pH + pOH = 14$

$\Rightarrow pH = 14 - pOH$

$\Rightarrow pOH = 14 - pH$

2. The strength of an acid depends on $[H^+]$ – ion present in it.

3. The strength of an base depends on $[OH^-]$ – ion present in it.

4. $pOH = -\log [OH^-]$

$pOH = \log [OH^-]^{-1}$

$pOH = \log \frac{1}{[OH^-]}$

i.e, $pOH \propto \frac{1}{[OH^-]}$

$\therefore pOH \uparrow$, $[OH^-] \downarrow$ & $pOH \downarrow$, $[OH^-] \uparrow$

▪ **Numerical based problem**

Q1. Find the pH of distilled water If $[H^+]$ in distilled water is 1×10^{-7} .

Ans Since given,

$$[H^+] = 1 \times 10^{-7}$$

$$\begin{aligned}\therefore \text{pH} &= -\log_{10} [H^+] \\ \Rightarrow \text{pH} &= -\log_{10} 1 \times 10^{-7} = -1 \times \log_{10}^{-7} \\ \Rightarrow \text{pH} &= -1 \times -7 \log_{10}^{10} \\ \Rightarrow \text{pH} &= 7 \times 1 \quad [\because \log_{10}^{10} = 1] \\ \Rightarrow \text{pH} &= 7\end{aligned}$$

So, the pH value of distilled water = 7.

→ Neutral

Q2. If the $[H^+]$ of Hcl is 1×10^{-1} then find the value of pH.

Ans

$$\begin{aligned}\therefore [H^+] \text{ of Hcl} &= 1 \times 10^{-1} \\ \therefore \text{pH} &= -\log_{10} [H^+] \\ &= -\log_{10}^{1 \times 10^{-1}} = -\log_{10}^{10^{-1}} \\ &= -1 \times -1 \log_{10}^{10} = 1 \times 1 \quad [\because \log_{10}^{10} = 1] \\ \therefore \text{pH of Hcl} &= 1\end{aligned}$$

Q3. If the pH value of a solution is 5 then find its pOH value.

Ans

$$\begin{aligned}\therefore \text{pH} &= 5 \\ \& \text{ pH} + \text{pOH} &= 14 \\ \Rightarrow \text{pOH} &= 14 - \text{pH} \\ \Rightarrow \text{pOH} &= 14 - 5 = 9\end{aligned}$$

Q4. If the pOH value of any solution is 8 then find its pH - value.

Ans

$$\begin{aligned}\therefore \text{pOH} &= 8 \\ \& \text{ pH} + \text{pOH} &= 14 \\ \Rightarrow \text{pH} &= 14 - \text{pOH} \\ \Rightarrow \text{pH} &= 14 - 8 = 6\end{aligned}$$

Special Note:

1.

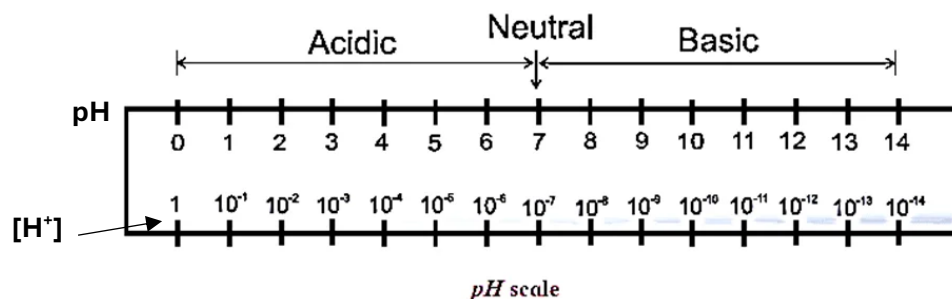
- i) The pH - value of neutral solution = 7.
- ii) The pH - value of acidic solution < 7.
- iii) The pH - value of basic solution > 7.

For example:-

- ✓ The pH - value of neutral soil and distilled water = 7.
- ✓ The pH - value of Hcl = 1.
- ✓ The pH - value of NaOH = 10.

2. pH values lies between 0 to 14 but

- ✓ If pH value $< 0 \rightarrow$ Extremely acidic.
- ✓ If pH value $> 14 \rightarrow$ Extremely basic.



Thus,

- i) $\text{pH} < 7 \rightarrow$ Acidic Solution
- ii) $\text{pH} = 7 \rightarrow$ Neutral Solution
- iii) $\text{pH} > 7 \rightarrow$ Basic Solution

▪ **Some substances and their pH – value**

Substance / Solution	pH – value
[Hcl] – Diluted Hcl	0 – 1
Gastric Juice	1.4
Lemon Juice	2.5
Vinegar (Sirka)	3.0
Wine	3.5
Vinegar	4.0
Tomato Juice	4.1
Coffee	5.0
Saliva (after meals)	5.8
Soft Drink & Urine	6.0
Milk	6.5
Pure Water & Soil	7
Saliva (Before meals)	7.4
Blood	7.4 – 7.5
Egg	7.8
Toothpaste	8.0
Baking Soda Solution	8.5
Washing Soda Solution	9.0
Milk of magnesia	10.5
Lime Water	11
Household Ammonia	11.6
Diluted Sodium Hydroxide	13.0
Concentrated NaOH	14

▪ **Importance of pH in everyday life**

1. To check the pH of water for the purpose survive plants, animals, Fish etc.
2. To check the pH of soil for the purpose of agriculture and amount of fertilizer required.

3. To check the pH of blood of our body for the purpose of checking disease. If pH differ then many disease play in role.
4. To check the pH of gastric juice for the purpose of stomach disease.
5. To check the pH of tooth decay for the purpose of amount of bacteria present in mouth produce acids.
6. Self-defense by animals and plants through chemical warfare.

▪ **The pH – value of Salt**

1. The pH – value of those salt which solution is formed by the strong acid and strong base are neutral. So, its pH value is 7.
2. The pH value of those salts which solution is formed by the strong acid & weak base are acidic. So, their pH value is less than 7.
3. The pH value of those salts which solution is formed by the weak acid and strong base are basic. So, their pH value is more than 7.

▪ **Common Salt (NaCl)**

- Sodium Chloride (NaCl) is also known as common salt.

▪ **Formation of Common Salt is laboratory**

- When aqueous solution of sodium hydroxide (NaOH) reacts with aqueous solution of HCl then Sodium Chloride (NaCl) and water is formed.



▪ **Widely Common Salt is formed by**

- Common Salt is obtained from sea water by the process of evaporation. This is done as follows. Sea water is trapped in large shallow pools and allowed to stand there. The sun's heat evaporate the water slowly and common salt is left behind.

Note:

1. Sea water is salty due to presence of rock salt.
2. Rock salt is also mined from the underground deposit like coal.

▪ **Uses of Common Salt**

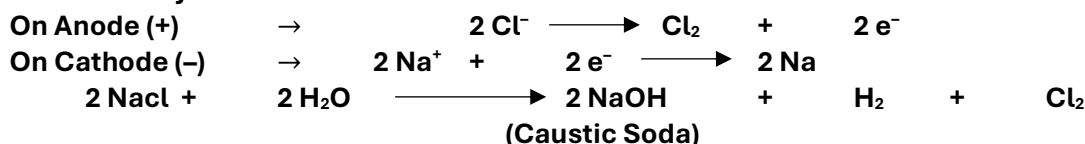
1. It is used in cooking food.
2. It is used as preservative in the meat, pickles etc.
3. It is used in the manufacturing of soap.
4. In making toothpaste.
5. In melting ice which collect on the road during winter in cold countries.
6. In making a large no. of useful chemicals in the industry like NaOH, Na_2CO_3 , NaHCO_3 , HCl etc.

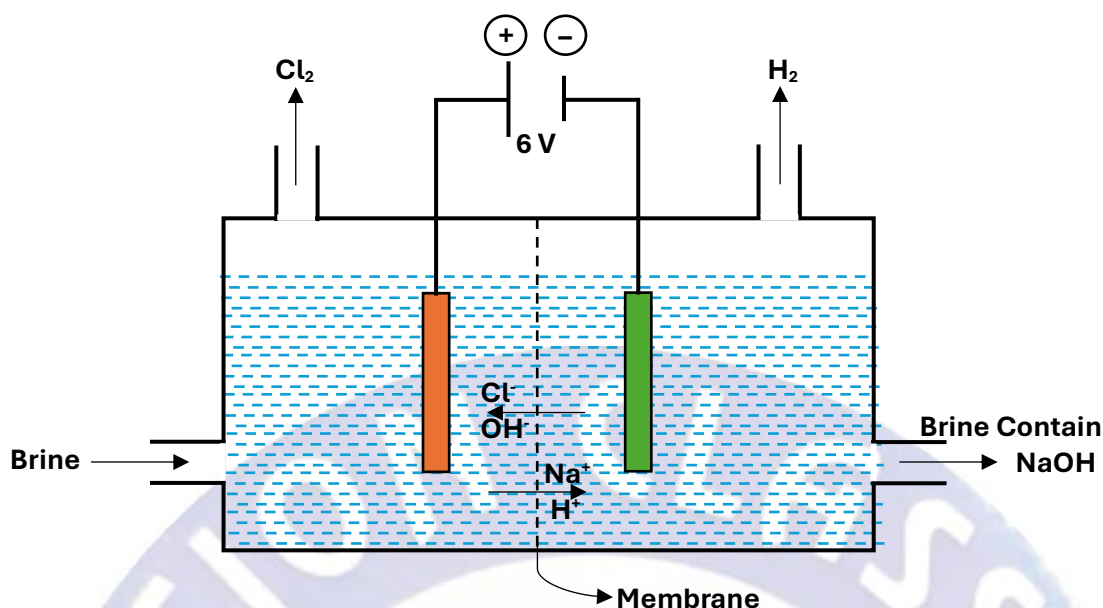
Famous Name

• **Caustic Soda / Sodium Hydroxide**

M.F = NaOH , pH = 10 [Base]

- Caustic Soda is prepared (Made up) by “chlor – alkali” process.
- In this process, The Saturated aqueous solution of sodium chloride, called Brine is decomposed by electrolytic decomposition, Cl^- becomes oxidized and given chlorine gas on anode (+) while Na^+ becomes reduced and converted sodium metal on Cathode (-). Now Sodium (Na) reacts with water to form NaOH.
- It is shown by its chemical reactions.





▪ Uses of Sodium Hydroxide

- To make soap and detergent.
- To make paper.
- As reagent in laboratory.
- To make synthetic cloth like rayon etc.
- To get aluminum from bauxite ores.
- To purify oil.
- To make dyes and bleaching powder.

▪ Uses of Chloride gas

- To bleach paper and cloths.
- To kill germs present in water.
- To make bleaching powder.
- To make hydrochloric acid.
- To make plastic, P.V.C etc.
- To make chloroform (CHCl_3) and paints.

▪ Uses of Hydrogen gas

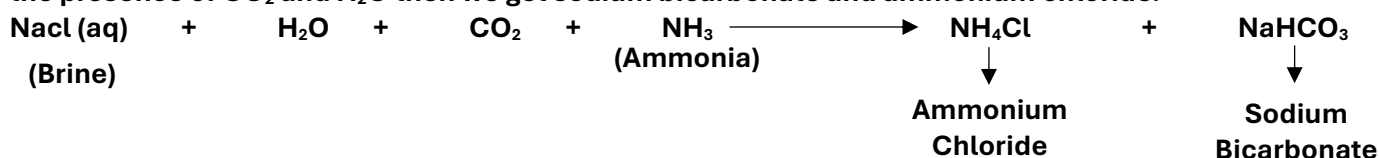
- It is used in hydrogenation of oil to obtain fats called vegetable ghee.
- In the formation of HCl .
- To make Ammonia (NH_3) by Haber method.
- To make methanol (CH_3OH).
- Liquid Hydrogen is used as fuel of rockets.

Famous name

• Baking Soda / Sodium Bicarbonate / Sodium Hydrogen Carbonate

M.F = NaHCO_3 , [Base]

- It is made by Ammonia – Soda method.
- In this method, The saturated aqueous solution of Sodium chloride called brine reacts with ammonia in the presence of CO_2 and H_2O then we get sodium bicarbonate and ammonium chloride.



▪ Properties of Baking Soda

- The aqueous solution of NaHCO_3 is alkali. So, the pH – value is more than 7.
- NaHCO_3 keep white crystal which is slightly soluble in water.
- It neutralize acid and release CO_2 .

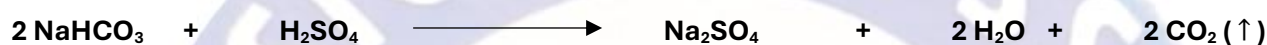


- When NaHCO_3 is heated we get Sodium Carbonate.

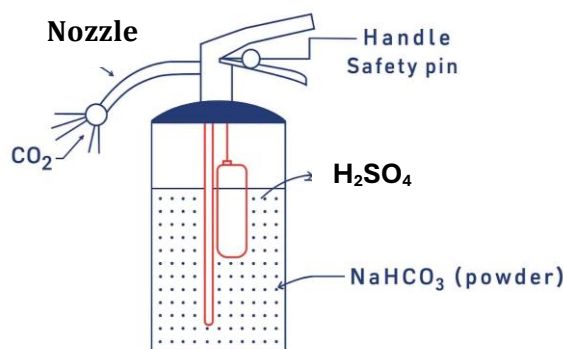


▪ Uses of Baking Soda

- To make baking powder which is used to makes cakes & bread.
- To reduces acidity of stomach.
- In fire extinguisher it is also used.



Produces



FIRE EXTINGUISHER

- In kitchen to make food fast & salt.

Note:

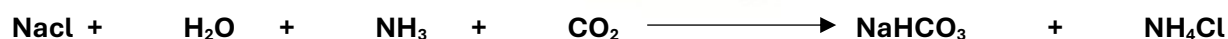
- Baking powder is a mixture of baking Soda and tartaric acid.

Famous name

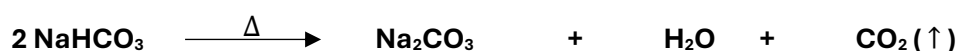
• Washing Soda / Sodium Carbonate

M.F = $\text{Na}_2\text{CO}_3 \cdot 10 \text{H}_2\text{O}$, Nature = Base

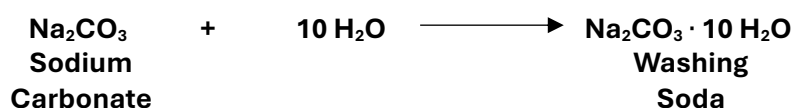
- It is made by ammonia Soda method.
- In this method, The saturated aqueous solution of sodium chloride called brine with ammonia and then carbon dioxide gas is passed into the liquid Sodium chloride is converted into Sodium bicarbonate.



- Now, NaHCO_3 is heated, then we get Na_2CO_3 .



- When crystal of sodium carbonate is washed we get Washing Soda.

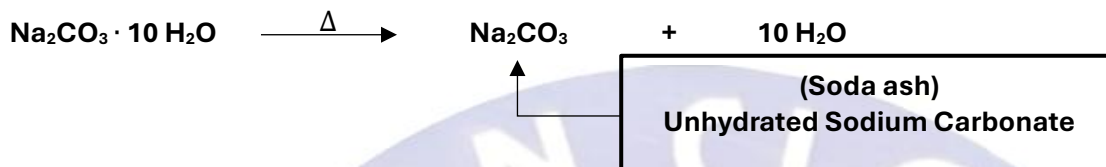


▪ **Properties of Washing Soda**

- The aqueous solution of Na_2CO_3 is alkali so, its pH – value is more than 7.
- Na_2CO_3 neutralise acid and give $\text{CO}_2 (\uparrow)$.



- When $\text{Na}_2\text{CO}_3 \cdot 10 \text{H}_2\text{O}$ heated fastly we get unhydrated Na_2CO_3 (Soda ash).



- When CO_2 gas is flowed in aqueous solution of $\text{Na}_2\text{CO}_3 \cdot 10 \text{H}_2\text{O}$. we get sodium bicarbonate.



▪ **Uses of Washing Soda**

- To wash clothes.
- As reagent in laboratory.
- To make glass, soap, paper etc.
- To remove hardness of water.
- To make sodium compound such as $\text{Ca}(\text{OH})_2$.

Famous name

- **Bleaching powder OR Calcium oxychloride**
 M.F = CaOCl_2 Ca (OCl) Cl

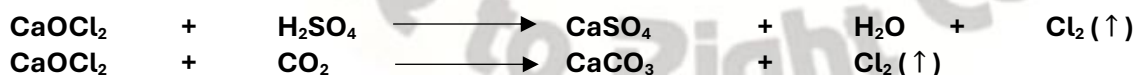
▪ **Method to make**

- When dead lime $\text{Ca}(\text{OH})_2$ is heated to 40°C and flowed to chlorine gas we get bleaching powder.

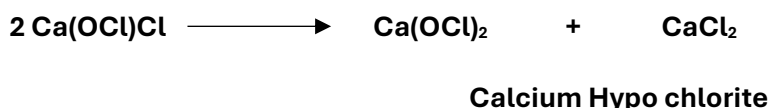


▪ **Properties of bleaching powder**

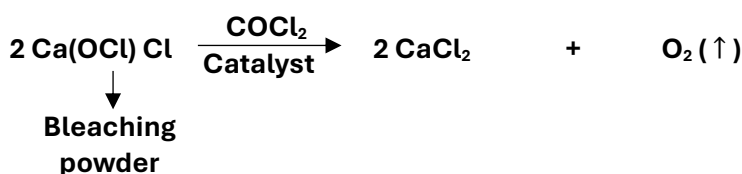
- It is a white powder from which smell of chlorine comes.
- When it is kept in air reacts with CO_2 and diluted acid, it release chlorine gas.



- When it is mixed in water then it forms calcium chloride and calcium hypo chlorite.



- In the presence of COCl_2 Solution bleaching powder loses its oxygen.



▪ Uses of Bleaching powder

- To kill germs.
- To bleach cloths & paper etc.
- To make chlorine and chloroform.

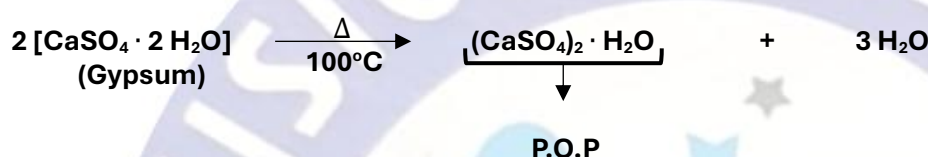
• Plaster of Paris (P.O.P) $\rightarrow$ $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$ OR Calcium Sulphate Hemihydrate = $(\text{CaSO}_4)_2 \cdot \text{H}_2\text{O}$

▪ Method of preparation

- When Gypsum is heated to $120^\circ\text{C}/100^\circ\text{C}$ fastly then plaster of paris is formed.



OR



Note:

- If the temperature is more than 120°C then unhydrated Calcium Sulphate is formed which is called dead burnt plaster.

▪ Properties of P.O.P

- It is a white powder.
- When it is mixed in water then it becomes hard and tiny holes forms in it fastly, this is because it returns in the form of Gypsum.



▪ Uses of Plaster of Paris (P.O.P)

- In making Statues.
- To close holes in laboratory.
- To join broken bones into surgery.
- It is used as fire – proof material.

• Gypsum $\rightarrow$ $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

▪ Method to make

- When water is mixed in plaster of paris then it becomes hard which is called Gypsum.

